

Tech Stack Decision Rationale

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1. DEFINED CONSTRAINTS & REQUIREMENTS

- **Budget Constraint:** Strictly \$0 (100% free hosting and deployment pipelines).
- **Developer Skill Level:** Intermediate Frontend & AI/ML Engineer (HTML, CSS, JavaScript, React/Next.js).
- **Portfolio Purpose:** A central directory hub showcasing AI sub-projects, ML research tools, brand kits, and content maps.
- **Display & Media Needs:** Direct sub-path hosting, live embedded demos, GitHub links, and clean UI navigation.
- **Dynamic Requirement:** Static architecture for zero maintenance overhead.

2. EVALUATED TECH STACK ALTERNATIVES

✓ **Selected Stack: Modular Vanilla HTML5/CSS3 Landing Directory + GitHub Pages**

- **How It's Built:** Custom semantic HTML5 layout with dark-mode CSS styling (`index.html` + `style.css`), hosting cloned sub-repository folders directly in sub-paths.
- **Hosting Platform:** GitHub Pages (100% Free).
- **Backend Required:** None (Purely static client-side rendering).
- **Key Trade-off:** Requires basic manual maintenance of HTML links when adding new modules, but offers complete design freedom and zero deployment failures.

✗ **Rejected Option A (The Simplest): Notion / Canvas No-Code Builders**

- **How Built / Hosted:** Drag-and-drop page creation hosted on free tiers.
- **Why Rejected:** Lacks custom CSS control, cannot host independent repository sub-directories natively, and restricts live interactive UI embeds.

✗ **Rejected Option B (The Most Powerful): Next.js Framework + Tailwind CSS + Vercel**

- **How Built / Hosted:** Full-stack SSR/SSG React framework deployed on Vercel free tier.

- **Why Rejected:** Unnecessary framework complexity, dependency updates, and build pipeline overhead for what is currently a static project hub.

3. PRESSURE-TEST & FINAL JUSTIFICATION

Can I finish in two weeks?

Yes. The directory framework, CSS design system, and sub-module folder organization are fully completed and tested in under 2 days.

Can I maintain this long-term?

Yes. Adding a new project simply requires cloning the repository into a sub-folder and pasting a small HTML card snippet into ``index.html``.

Does it show my work effectively?

Yes. It provides a clean, professional dark-mode UI hub (``all-projects-hub``) that immediately routes evaluators to live sub-projects and source code repositories.